

AI Agent Workflow Design

Sequential vs Parallel vs Router • Nodes • State Graph Design

Quick Reference: Three Core Workflow Patterns

Pattern	When to Use	Typical Shape	Key Question
Sequential	A step depends on the output of the previous step.	$A \rightarrow B \rightarrow C \rightarrow D$	Does each step require the previous step's result?
Parallel	Multiple independent tasks can run at the same time and later merge.	$A \rightarrow \{B, C, D\} \rightarrow E$	Can these tasks execute independently on the same input?
Router	The next path depends on a classification, condition, risk level, request type, or business rule.	$A \rightarrow \text{Router} \rightarrow B/C/D$	Which path should this case follow?

Deliverable Expected for Every Scenario

- A. Workflow pattern selected
- B. Business justification
- C. Node list with node responsibilities
- D. State definition / state variables
- E. Edge definition
- F. StateGraph
- G. Failure / exception paths
- H. Explanation of why alternative workflow patterns are not optimal

Healthcare Domain Scenarios

Healthcare workflows often combine deterministic routing, independent analysis, validation, and human escalation. Learners should distinguish clinical/business decisions from independent analysis tasks and should avoid creating unnecessary sequential dependencies.

Scenario H1 — Patient Intake and Clinical Triage

Domain: Healthcare | Difficulty: Advanced

A hospital wants to build an AI-assisted patient intake workflow for its outpatient department.

A patient submits symptoms, age, gender, known conditions, current medications, recent reports, and free-text concerns through a digital intake form.

The system must first validate that mandatory intake fields are present. It then needs to determine the broad complaint category: cardiology, respiratory, gastrointestinal, neurological, orthopaedic, general medicine, or other.

Once the category is identified, the case should follow an appropriate triage path. A high-risk case should be escalated for immediate human review. A low-risk case can proceed to appointment prioritization. The workflow should also generate a concise intake summary for the clinician.

Important constraint: the system must not make a final medical diagnosis. It is an intake, prioritization, and summarization workflow.

Learner Tasks

- Should the overall workflow be Sequential, Parallel, Router, or a combination?
- Identify the node that determines the complaint category and explain why it should be a routing node.
- Design separate paths for high-risk, routine, and incomplete cases.
- Identify any tasks that can run in parallel after the patient information has been validated.
- Define the State fields required to preserve patient input, category, risk assessment, missing fields, summary, and routing decision.
- Draw the complete StateGraph including START, END, conditional edges, and convergence points.
- Explain how the workflow should behave when confidence in the category or risk assessment is low.
- Explain why a purely sequential design would create unnecessary latency.

Scenario H2 — Medical Record Review and Pre-Consultation Brief

Domain: Healthcare | Difficulty: Advanced

A specialist clinic receives a patient before consultation. The system has access to several documents: previous consultation notes, laboratory reports, imaging reports, medication history, and recent hospital discharge summaries.

The clinic wants an AI workflow that creates a pre-consultation brief. The brief should contain: (1) a chronological medical timeline, (2) abnormal laboratory findings, (3) medication changes, (4) important findings from imaging reports, and (5) unresolved follow-up items.

Each of these analyses can be performed independently once the source documents have been retrieved and normalized. After the independent analyses finish, another node must consolidate them into one

clinician-friendly brief. A validation step must check whether the final brief contains unsupported claims or contradictions.

Learner Tasks

- Which workflow pattern best represents the core analytical stage?
- Identify the independent branches and the final merge/consolidation node.
- Explain what data must be placed in shared state before parallel processing begins.
- Design a validation node after consolidation.
- What happens if one branch fails while the other branches succeed? Define a sensible state representation for partial completion.
- Draw the StateGraph and clearly show the fan-out and fan-in.
- Explain why executing all five analytical tasks sequentially is inferior.

Scenario H3 — Insurance Pre-Authorization Request

Domain: Healthcare / Health Insurance | Difficulty: Advanced

A health insurer receives a hospital request for pre-authorization. The request contains patient details, diagnosis/procedure information, estimated cost, clinical notes, supporting documents, and provider details.

The workflow must first validate the request. It then needs to identify the request type and route it to the appropriate policy pathway. For example, routine procedures may follow an automated policy check, while expensive or unusual procedures may require enhanced review. Missing documentation must be routed to a documentation-request path.

For valid requests, several independent checks are required: policy eligibility, document completeness, provider eligibility, and benefit coverage. The final decision-support node should combine the results and recommend approve, request more information, or send to human review. The workflow must not make an irreversible coverage decision without the organization's required controls.

Learner Tasks

- Is this a single pattern or a hybrid pattern? Identify each pattern used and where.
- Design the router and list its routing conditions.
- Identify which checks can execute in parallel.
- Design the merge node that converts multiple check results into one decision-support package.
- Define state fields for eligibility, coverage, provider status, missing documents, risk flags, and recommendation.
- Design exception paths for inconsistent evidence and low-confidence extraction.
- Create StateGraph.

Finance Domain Scenarios

Finance workflows commonly involve deterministic policy rules, risk segmentation, independent checks, document analysis, and human approval. Learners should make a clear distinction between routing logic and analytical work that can safely execute concurrently.

Scenario F1 — Loan Application Credit Assessment

Domain: Finance / Lending | **Difficulty:** Advanced

A retail bank receives a personal loan application containing applicant information, income documents, bank statements, credit history, existing liabilities, requested loan amount, and purpose.

The workflow first validates the application and extracts structured information from the submitted documents. It then calculates affordability indicators and analyzes credit history. These analytical tasks use the same normalized applicant data and can be performed independently.

A policy/risk classification stage then determines whether the application is low risk, medium risk, high risk, or requires additional information. Low-risk cases can move to an automated recommendation path, medium-risk cases require additional checks, and high-risk cases must be escalated for human review. The final node produces an explainable recommendation package rather than an unconditional approval.

Learner Tasks

- Identify the sequential portion of the workflow.
- Identify the parallel analytical branches.
- Where should the Router be placed, and what inputs should it use?
- Define at least four possible routing outcomes.
- Define the State object and explain which fields are raw inputs, derived features, decisions, and audit information.
- Design a merge node for affordability and credit-history results.
- Explain what should happen if income extraction succeeds but bank-statement extraction fails.
- Create StateGraph.

Scenario F2 — Fraud Detection for a Digital Payment

Domain: Finance / Payments | **Difficulty:** Advanced

A payment platform receives a transaction containing amount, merchant, customer, device, location, time, historical behavior, and recent transaction activity.

The system must quickly determine whether the transaction should be allowed, challenged, held for review, or declined. Several independent signals can be evaluated at the same time: velocity checks, device-risk analysis, location anomaly analysis, merchant-risk analysis, and customer behavior analysis.

The outputs are then combined into a fraud-risk score. A policy router maps the resulting risk band and selected signals to an operational action. A high-risk transaction may be held, while a suspicious but lower-confidence transaction may trigger additional customer verification.

Learner Tasks

- Why is Parallel a strong fit for the signal-generation stage?
- Design the fan-out branches and define the output of each branch.
- Design the merge/scoring node.
- Design the Router after scoring and list possible actions.
- Define the minimum State needed to pass information between branches and the router.
- Explain how you would handle a branch timeout without blocking the entire workflow indefinitely.
- Create StateGraph

Scenario F3 — Investment Research Request Router

Domain: Finance / Investment Research | Difficulty: Advanced

An investment research desk receives requests from analysts. A request might concern an individual company, a sector, a macroeconomic theme, an earnings event, or a portfolio-level question.

The workflow should first classify the request. Based on the request type, it should route the request to a specialized research path. A company-analysis path may require financial-statement analysis and recent-news review. A macro path may require economic indicators and policy/news research. A portfolio path may require exposure analysis and risk concentration analysis.

The workflow should finish with a synthesis node that produces a standardized research response with sources, assumptions, key findings, risks, and unanswered questions.

Learner Tasks

- Why is Router the dominant pattern in this scenario?
- Design the request-classification node and routing criteria.
- For one selected route, identify tasks that can execute in parallel.
- Explain how the State schema can remain consistent across routes even though different branches use different data.
- Design a final synthesis node that can consume route-specific results.
- Define a fallback path for ambiguous requests.
- Create StateGraph.

Telecom Domain Scenarios

Telecom workflows frequently combine event classification, network/service diagnostics, independent health checks, customer segmentation, and escalation. Timing and convergence are important because unnecessary sequential processing can increase operational response time.

Scenario T1 — Network Incident Diagnosis and Escalation

Domain: Telecom / Network Operations | Difficulty: Advanced

A telecom operator detects a sudden increase in dropped calls in a geographic region. The operations platform receives alarms, KPI deviations, affected cell sites, recent configuration changes, transport-network events, and customer complaints.

The workflow should first determine the incident type and severity. It then needs to investigate multiple independent dimensions: radio access network health, transport/backhaul health, configuration changes, power/environmental events, and customer-impact patterns.

Once those analyses complete, the workflow should correlate the evidence and determine whether the incident is likely isolated to a cell site, a cluster, a transport segment, or a broader regional problem. The incident should then be routed to the correct operational team and severity process.

Learner Tasks

- Identify the sequential setup steps.
- Design the parallel diagnostic branches.
- Identify the correlation/merge node.
- Design the Router that maps diagnosis and severity to the responsible operational path.
- Define State fields for incident metadata, diagnostic findings, evidence, confidence, impact, severity, and assignment.
- Design a fallback path when diagnostic branches disagree.
- Explain how you would prevent one slow diagnostic branch from making the whole workflow unusable.
- Create StateGraph.

Scenario T2 — Customer Complaint Resolution

Domain: Telecom / Customer Service | Difficulty: Advanced

A customer submits a complaint such as poor network coverage, slow data speed, unexpected billing, SIM/eSIM problems, service activation failure, or roaming issues.

The workflow first validates the customer and retrieves the relevant account/service context. A classification node identifies the complaint category and routes it to the appropriate troubleshooting path.

For a network-quality complaint, the workflow may independently check recent service outages, location coverage, device compatibility, and recent network performance. For a billing complaint, it may instead inspect invoices, plan configuration, usage records, and applicable discounts.

The final response should be generated from the evidence collected and should indicate whether the issue can be resolved automatically, requires customer action, or needs escalation to a specialist.

Learner Tasks

- Why is Router the key pattern at the complaint-classification stage?
- Show how one route can itself contain a Parallel pattern.
- Design at least five complaint routes.
- Define a common State schema that works across different complaint types.
- Design the final response node so it can consume route-specific evidence.
- Add an escalation path for unresolved or low-confidence cases.
- Create StateGraph for the overall workflow and expand one route in detail.

Scenario T3 — Telecom Churn Risk and Retention Action

Domain: Telecom / Customer Analytics | Difficulty: Advanced

A telecom company wants to identify customers at risk of churn and determine an appropriate retention action.

The workflow receives customer profile, tenure, plan, usage patterns, billing behavior, service quality metrics, support interactions, and recent changes in behavior.

Multiple analytical modules independently assess different churn signals. The outputs are merged into a churn-risk assessment. Based on risk level and customer segment, the workflow routes the customer to a suitable action: no intervention, personalized offer, service-quality intervention, billing support, or human retention specialist.

The company wants an explainable record of why a customer was classified as high risk and why a specific retention action was selected.

Learner Tasks

- Identify the parallel analytical branches.
- Design the merge node that creates the overall churn-risk profile.
- Design the Router using churn risk, customer value, and issue type.
- Identify what should be stored in State for explainability.
- Design the final action-selection node.
- Explain why the retention-action logic should not simply be another parallel branch.
- Create StateGraph.